Additional Guidance/Help for Assignment 3.2

1. Given electrochemical device is a reversible (fuel cell/electrolyzer) cell. When the current is negative (I<0), the device operates in the electrolyzer mode, the oxygen electrode becomes the anode:

$$\frac{1}{2}O_2 + 2e^- \leftarrow O^{2-}$$

2. We are given the cell resistance (which we will define as area specific resistance, R_{ASR}) in Module 4. We can first calculate the IR corrected potential (also IR free potential, cell potential compensating for ohmic losses):

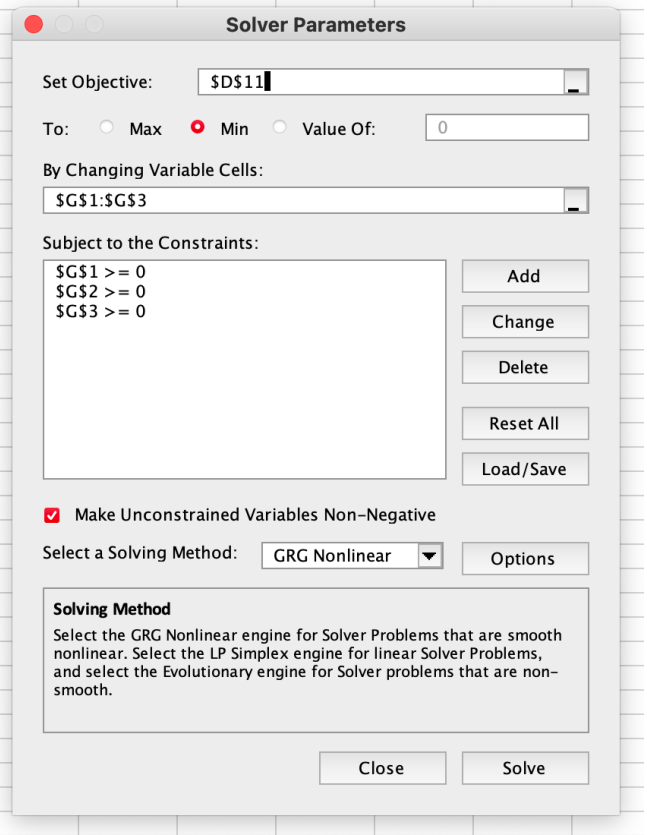
$$V_{IRfree} = V_{cell} + IR_{ASR}$$

Note that with this correction, cell potential gets closer to the equilibrium potential both in electrolyzer ($I_{cell} < 0$)and fuel cell mode ($I_{cell} > 0$).
Further, if hydrogen electrode losses can be negligible, the oxygen electrode overpotential can then be calculated as:

$$\eta_{Ox} = V_{IRfree} - E_{eq}$$

Note that $\eta_{Ox} > 0$ in the electrolyzer mode as the oxygen electrode becomes the anode ($1/2O_2 + 2e^- \leftarrow O^{2-}$); and $\eta_{Ox} < 0$ in the fuel cell mode ($1/2O_2 + 2e^- \rightarrow O^{2-}$)

3. The oxygen electrode current is the inverse of the given cell current as in the electrolyzer mode cell current, $I_{cell} < 0$ but since oxygen electrode is anode, it will have $I_{Ox} > 0$; and in the fuel cell mode, $I_{cell} > 0$ but since oxygen electrode is cathode, it will have $I_{Ox} < 0$

4. We then fit η_{Ox}, I_{cell} to B-V equation with the unknowns of i_o, α_a, α_c. My approach is to define these unknowns as cells in Excel and define the objective function as the sum of squared residuals, $\sum(I_{cell} - I_{BV})$. Column C calculates the oxygen electrode overpotential from the oxygen electrode current using the Butler-Volmer equation with the parameters listed in Column G. See the layout below:

	A	B	C	D	E	F	G	H
1	I_Ox	eta_Ox (exp)	eta_Ox (B-V)	R^2 (ColumnC- Column D)^2		i_o	4056.87675	A/m2
2	6981	0.4638273	7068.270189	7616.085841		alpha_a	0.14520293	
3	4871	0.3327643	4785.162178	7368.13161		alpha_c	0.12873774	
4	2946	0.1976618	2718.057174	51957.93194		F/RT	11.9271571	1/V
5	967	0.0584211	780.0194585	34961.7229				
6	-930	-0.079069	-1042.86182	12737.79001				
7	-2799	-0.2167467	-2871.65706	5279.048377				
8	-4753	-0.3561549	-4820.25103	4522.700945				
9	-6836	-0.4923988	-6911.48331	5697.730416				
10	-9328	-0.6275024	-9264.19709	4070.811819				
11				134211.9539